

HISTOLOGY

1. Concerning the *teeth and associated tissues*,
a) Dentin is more calcified than enamel. b) Pulp is dense connective tissue.
c) Enamel is replaced in the root by cementum. d) Odontoblasts make enamel.
e) Periodontal membrane is loose connective tissue.

2. Which of the following types of *lingual papillae* lack *taste buds*?
a) circumvallate b) filiform c) foliate d) fungiform e) none

3. Regarding the *stomach*, which statement is *true*?
a) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.
b) Has its gastric glands best developed in the pylorus than the other regions.
c) forms temporary folds called plicae circulares when the stomach is empty.
d) Has its muscularis externa made by skeletal muscle fibres.
e) None of the above.

4. Pernicious anemia can result from insufficient intrinsic factor secretion by which of the following cells?
a) Paneth cells b) enterocytes c) ameloblasts
d) oxyntic (parietal) cells e) zymogenic (chief) cells

5. Concerning the *stomach*, which statement is *true*?
a. It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.
b. Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.
c. Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.
d. Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.
e. The folds of the empty stomach are called rugae.

6. Which one of the following marks best the *ileum* from the *duodenum* and *jejunum*?
a) Long and broad leaf like villi. b) Absence of teniae coli. c) Peyer's patches.
d) Plicae circulares. e) Well developed intestinal glands.

7. Which of the following organs characteristically contains the most *goblet cells* in the mucosa covering its *villi*?
a) duodenum b) ileum c) colon d) cecum e) stomach

8. *Vermiform appendix*,
a) has few lymphoid nodules in the lamina propria and submucosa.
b) is a blind-ended evagination of the ileum. c) has relatively shallow intestinal glands.
d) longitudinal smooth muscle gathers into its teniae coli. e) contains short villi.

9. With regard to the *small intestine*, which statement is *true*?
- The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.
 - Crypts of Lieberkühn are lymphatic capillaries [lacteals] within a villus of the small intestine.
 - Microvilli that seen on the columnar epithelial cells in the small intestine form the brush border or striated border.
 - The plicae circulares are projections with a core of mucosa.
 - Villi are finger like projections of submucosa.
10. Diarrhea may result if which of the following organs fails to carry out its primary role in *absorbing water from the feces*?
- anal canal
 - colon
 - cecum
 - small intestine
 - stomach
11. With regard to the *large intestine*, which statement is *false*?
- Paneth cells of the submucosal glands house bacteria that produce vitamin B₁₂ and vitamin K.
 - Lacks villi and contains a disproportionately large number of goblet cells.
 - The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.
 - The muscularis externa has three thick bands of muscle called the tenia coli.
 - The large intestine is shorter than the small intestine but has a wide lumen.
12. *Hepatic sinusoids*,
- are bordered directly by hepatocytes.
 - contents flow toward the portal triads.
 - are surrounded by the space of Disse.
 - contents empty into the canal of Herring.
 - lumen is entirely sealed by junctional complexes.
13. Which is *not* the *function of the liver*?
- Detoxification/removal of drugs and alcohol.
 - Storage of calcium and bile.
 - Storage of glycogen, fat soluble vitamins, iron and cholesterol.
 - Production of plasma proteins and lipoproteins.
 - Fetal RBC production
14. With regard to the *pancreas*, which statement is *false*?
- Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
 - The myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
 - The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
 - Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
 - It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.
15. Concerning the *GI tract*, select the statement which is *not correct*.
- The mucosa of the gallbladder is made of simple columnar epithelium.
 - The masticatory mucosa is located on the hard palate and gingiva.
 - The adventitia consists of only a single layer of epithelial cells and underlying connective

tissue.

- d. The muscularis externa consists of Auerbach's plexus.
- e. Brünner's glands in submucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone.

16. In the *digestive system*, select the statement which is *false*.

- a) Parotid glands produce purely serous secretion in the adult.
- b) Paneth cell secretes lysozyme.
- c) the exocrine pancreas contain centroacinar cells.
- d) the hepatic portal vein carry blood rich in bilirubin.
- e) cholecystokinin stimulates the relaxation of smooth muscle in the gallbladder.

17. Which organ in the GIT has a *mucosa lined by nonkeratinized stratified squamous epithelium*?

- a. Esophagus b. Stomach c. Small intestine d. Large intestine e. Rectum

18. What type of cells are found in the olfactory mucosa?

- a. Basal cells b. Brush cells c. Olfactory cells d. Sustentacular cells
- e. All of the above

19. Which of the following *components of the respiratory tract* wall increase(s) in amount from trachea to alveoli?

- a) cilia b) elastic fibers c) smooth muscle d) cartilage e) goblet cells

20. What type of epithelium is found in the *respiratory mucosa* of man?

- a. Non-ciliated pseudostratified columnar epithelium with goblet cells
- b. Ciliated pseudostratified columnar epithelium with goblet cells c. Transitional epithelium
- d. Simple columnar epithelium e. Stratified squamous epithelium

21. Which cell is a *respiratory macrophage*?

- a. Kupffer cell b. Histocyte c. Dust cell or alveolar macrophage
- d. Langerhans cell e. Microglia

22. *Bronchioles*:

- a) Have well developed cartilages b) Are lined by stratified squamous epithelium
- c) Have no muscle d) Possess Clara cells e) Have mucous producing glands

23. The *visceral pleura*,

- a) can be described as an adventitia. b) is devoid of lymphatic vessels.
- c) includes a layer of simple squamous epithelium. d) lacks elastic fibers.
- e) covers the outer wall of the pleural cavity.

24. *Pulmonary surfactant*,

- a) is secreted by type I pneumocytes. b) may promote the growth of bacteria.
- c) consists of a monolayer of phospholipids overlying an aqueous hypophase.
- d) prevents alveolar collapse by increasing surface tension.

e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

25. *Pneumocytes type I:*

- a) are found in large numbers in the terminal bronchioles.
- b) Could give rise to pneumocytes type II.
- c) Produce surfactant.
- d) Have large round centrally located nuclei.
- e) Line more surface area of the alveolar wall than do the alveolocytes type II.

26. Select the *incorrect* statement in the *respiratory system*.

- a) The vestibule of the nose is lined by stratified squamous epithelium.
- b) The paranasal sinuses are lined by ciliated pseudostratified epithelium with goblet cells.
- c) Larynx is partly lined by nonkeratinized stratified squamous epithelium.
- d) A smoker's respiratory epithelium typically develops more goblet cells.
- e) The lamina propria contains mucous glands at the level of the alveolar ducts.

27. Which of the following is not a *function of the kidney*?

- a. Erythropoietin production
- b. Vitamin D modification
- c. Acid base balance
- d. Aldosterone production
- e. Renin production

28. What vessels arise from the *efferent arterioles*?

- a. Afferent arteriole
- b. Efferent arteriole
- c. Capillary tuft
- d. Peritubular capillaries
- e. Vasa recta

29. Which of the following is *not* the function of *mesangial cells*?

- a. synthesis of extracellular matrix.
- b. endocytosis and contractility.
- c. maintenance of acid-base balance in the blood.
- d. structural support to the glomerulus.
- e. increase blood flow and the effective surface area available for filtration.

30. Select the *correct* statement pertaining the *morphology of the kidney*.

- a. The distal convoluted tubules exhibit brush borders and apical canaliculi.
- b. The renal pyramid and its associated cortex is referred to as the renal lobe.
- c. Macula densa cells are located in the urinary pole of the nephrons.
- d. Renin is secreted from the visceral layer of Bowman's capsule referred to as area cribrosa.
- e. Interstitial cells in the renal medulla secrete erythropoietin through the filtration slits.

31. In the *urinary system*, which statement is *true*?

- a) The proximal convoluted tubule lacks a brush border.
- b) The cortical collecting tubules are located primarily at the apex of the renal pyramids and are lined by pseudostratified ciliated columnar cells.
- c) Extraglomerular mesangial cells are tall, narrow, closely packed epithelial cells of the distal tubule.
- d) Endothelial cells of glomerular capillaries are of non-fenestrated varieties.
- d) None of the above.

32. *Testis* has the following coverings *except*:

- a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis
e. All of them

33. In the *testis*, which statement is *true*?

- a. Interstitial cells (of Leydig) produce testicular fluid.
b. Sertoli cells are directly involved in spermatogenesis.
c. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
e. During spermiogenesis, the spermatids quickly undergo the first meiotic division to give spermatozoa.

34. In the *male reproductive system*, which structure is *not* lined by *pseudostratified epithelium*?

- a. vas deferens. b. seminal vesicles. c. epididymis
d. seminiferous tubules e. None of them

35. Which of the following statements is *false* in the *male reproductive system*?

- a. Sertoli cells coagulate semen and form a vaginal plug.
b. The subcutaneous tissue in penis is devoid of fat.
c. The secretion of seminal vesicle is particularly rich in fructose.
d. The largest male accessory gland, the prostate gland, is composed of compound tubuloalveolar glands found in fibromuscular [fibroelastic] stroma.
e. Cowper's glands [bulbourethral glands] are homologous to the greater vestibular glands in the female.

36. When does a *secondary oocyte* complete its second meiotic division to become a mature ovum?

- a) at ovulation b) before ovulation c) at fertilization
d) at puberty e) before birth

37. Which of the following is *true* of the *ampulla* of the *uterine tube* but *not* of its *isthmus*?

- a) Contains large mucosal folds that are extensively branched.
b) Contains smooth muscle in its walls. c) Is covered by serosa derived from peritoneum.
d) Contains ciliated epithelial cells. e) Contains mucus-secreting peg cells.

38. Which of the following statements is *true* about the *female reproductive system*?

- a. The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
e. Nonciliated secretory cells [peg cells] present in the lining epithelium of uterine tube synthesise and store glycogen.

39. Which of the following statements is *true* in the *female reproductive system*?

- a. Mammary gland is a modified apocrine sweat gland.

- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. Granulosa cells that are adherent to the zona pellucida form corona radiata.
- d. The basal layer of the endometrium is the thick smooth muscle tunic of the uterus.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the posterior vaginal fornix.

40. With respect to the *zona pellucida*, which statement is false?

- a. is composed of sulfated glycoproteins.
- b. prevents polyspermy
- c. prevents premature implantation
- d. functions as a porous filter
- e. arises by delamination from the embryoblast.

EMBRYOLOGY

1. The proximal convoluted tubules of the definitive adult kidney are derived from
 - a) ureteric bud
 - b) metanephric vesicle
 - c) metanephric duct
 - d) pronephric tubules
 - e) metanephric tubules

2. Concerning the development of the *urinary system*, which statement is *false*?
 - a) Urinary system begins to develop before the genital system.
 - b) The metanephric diverticulum is the primordium of the ureter.
 - c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
 - d) Differentiation of the nephrons depends upon induction by the collecting tubules.
 - e) The kidneys attain their definitive adult position toward the end of the fetal period.

3. Concerning the *development of the urinary bladder and urethra*, which statement is *false*?
 - a) The female urethra is derived from endoderm of the urogenital sinus.
 - b) In adults, the urachus is represented by the median umbilical ligament.
 - c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 - d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
 - e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

4. With regard to the *congenital anomalies of the urinary system*, which statement is *false*?
 - a) Accessory renal arteries are end arteries.
 - b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
 - c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
 - d) Unilateral renal agenesis is more common in males than females.
 - e) None of the above.

5. During the *development of the Urinary System*, which statement is *true*?
 - a) The urorectal septum divides the urachus from the allantois.
 - b) Bilateral renal agenesis is associated with oligohydramnios.
 - c) The podocytes of Bowman’s capsule are derived from endoderm and ectoderm.
 - d) Urachal cyst is a remnant of the cloaca.
 - e) Urine production begins shortly after birth.

6. Concerning the development of the urinary System, which statement is *true*?
 - a) Collecting duct is the adult derivative of the proximal convoluted duct.
 - b) The metanephros become functional just before birth.
 - c) The podocytes of the Bowman’s capsule are derived from neural crest cells.
 - d) The transitional epithelium lining the ureter is derived from endoderm and neural crest cells.
 - e) None of the above.

7. A newborn baby presents with fluid draining from the umbilicus onto the skin. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

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|-----------------|---------------------|-----------------------------|
| a) Urachal cyst | b) Urachal fistula | c) Exstrophy of the bladder |
| d) Bifid ureter | e) Horseshoe kidney | |

8. With regard to the development of the *thyroid gland*, which statement is *false*?

- a) The thyroid primordium consists of a solid mass of endodermal cells.
- b) The thyroid gland is connected to the tongue by a narrow tube, the thyroglossal duct
- c) The proximal opening of the thyroglossal duct persists as a small pit in the dorsum the tongue-the foramen cecum.
- d) After birth, the levels of fetal thyroid -stimulating hormone and thyroxine begin to increase, reaching adult levels during puberty.
- d) The parafollicular cells in the thyroid gland are derived from the ultimopharyngeal bodies.

9. Concerning the development of the *suprarenal gland*, which statement is *true*?

- a) The cortex develops from endoderm and the medulla differentiates from intermediate mesoderm.
- b) The zona reticularis and zona fasciculata are present at birth.
- c) The fetal medulla produces steroid precursors that are used by the placenta for the synthesis of estrogen.
- d) Cells in the fetal cortex do not proliferate; they give rise to the thin, poorly defined zona glomerulosa.
- e) None of the above.

10. With regard to the development of the *endocrine system*, which of the following statements is *false*?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
- b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
- c) the thyroid gland is the first endocrine gland to develop in the embryo.
- d) the pituitary gland is purely ectodermal in origin.
- e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

11. A structure found within the *adult female pelvis* formed from the *gubernaculum* is

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|--|-----------------------------------|------------------------------|
| a) broad ligament | b) superior ligament of the ovary | c) medial umbilical ligament |
| <u>d) round ligament of the uterus</u> | e) median umbilical ligament | |

12. The *labia minora* arise embryologically from

- | | | |
|----------------------------|---------------------------|----------------------|
| a) phallus | b) labioscrotal swellings | c) sinovaginal bulbs |
| <u>d) urogenital folds</u> | e) paramesonephric duct | |

13. The *uterine tubes* of the adult female are derived embryologically from

- | | |
|---------------------|------------------------|
| a) mesonephric duct | b) mesonephric tubules |
|---------------------|------------------------|

c) paramesonephric duct d) paramesonephric tubules e) uterovaginal primordium

14. In the *male*, *failure of urethral folds to fuse* completely results in

- a) hypospadias b) epispadias c) cryptorchidism
d) hydrocele e) congenital inguinal hernia

15. The *epoophoron* and *paroophoron*, which are important as cysts, arise from:

- a) The Mullarian ducts b) The paramesonephric ducts
c) The mesonephric ducts d) Leydig hormones e) Wolffian hormones

16. With regard to the *development of gonads*, which statement is *false*?

- a) The surface epithelium of the female gonad develops into oogonia.
b) Sex of an embryo is determined genetically at the time of fertilization.
c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
d) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.
e) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.

17. Concerning the *development of the genital system*, one statement is false?

- a) Bilateral cryptorchidism results in sterility.
b) Hypospadias is constantly associated with exstrophy of the urinary bladder.
c) The fetal testis produces antimüllerian hormone by Sertoli cells at 6-7 weeks.
d) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
e) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or the labia majora.

18. With regard to the *vestigial structures* derived from embryonic *genital ducts*, which statement is *true*?

- a) The prostatic utericle is homologous to the vagina.
b) The cranial end of mesonephric duct in males may persist as paradidymis.
c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

. Which statement is true about *female reproductive system*?

- a) the basal layer of endometrium is sloughed off during each menstruation.
b) Liquor folliculi stimulates the myoepithelial cells of the mature follicle to rupture.
c) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
d) Granulosa cells that are adherent to the zona pellucida form corona radiata.
e) None of the above.

- . Concerning the *development of the genital system*, which statement is false?
- a) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the mesonephric ducts.
 - b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
 - c) Interstitial cells (of Leydg) begin to secrete androgenic hormones by the end of the third trimester.
 - d) The uterine tubes, uterus, cervix, and superior part of the vagina all arise from the paramesonephric ducts.
 - e) The lower portion of the vagina is derived from urogenital sinus.